04-08-2026

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Experiment 3:

AIM: Study of Python Libraries for ML application such as Pandas and Matplotlib

Implementation of Pandas Library:

PROGRAM:

```
import pandas as pd
s=pd.Series(["ironman","captian","thor","hulk"])
print(s)
print(len(s))
print(type(s))
s=pd.Series(["ironman","captian","thor","hulk"],index=[3,4,1,2])
print(s)
print(len(s))
print(type(s))
fruit=["Mango","Kiwi","Apple","Cherry"]
cost=[200,300,400,100]
sales=pd.Series(cost,index=fruit)
print(sales)
```

Output:

```
0    ironman
1    captian
2       thor
3       hulk
dtype: object
4
<class 'pandas.core.series.Series'>
3    ironman
4    captian
1       thor
2       hulk
dtype: object
4
<class 'pandas.core.series.Series'>
Mango     200
Kiwi      300
Apple     400
Cherry    100
dtype: int64
```

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Data Frames using Pandas Library:

PROGRAM:

```
EmployeeData={'ID':[10,20,30,40],'Name':["ironman","captian","thor","hulk"], 'Age':[20,30,40,50], 'City':[ 'New York', 'Los Angeles', ' Chicago', 'San Antonio']}
EmpDB=pd.DataFrame(EmployeeData)
print(EmpDB)
EmpDB=pd.DataFrame(EmployeeData,index=[1,2,3,4])
print(EmpDB)
```

Output:

```
   ID     Name  Age        City
0  10  ironman   20    New York
1  20  captian   30  Los Angeles
2  30     thor   40     Chicago
3  40     hulk   50  San Antonio
   ID     Name  Age        City
1  10  ironman   20    New York
2  20  captian   30  Los Angeles
3  30     thor   40     Chicago
4  40     hulk   50  San Antonio
```

Empty Data Frames using Pandas Library:

PROGRAM:

```
df=pd.DataFrame()
print(df)
print(type(df))
CD=['id','Emp','Sal','Exp']
df=pd.DataFrame(columns=CD)
print(df)
new_row=pd.DataFrame([{"id":101,"Emp":"Manager","Sal":100000,"Exp":3}])
df=pd.concat([df,new_row],ignore_index=True)
new_row=pd.DataFrame([{"id":102,"Emp":"SalesPerson","Sal":60000,"Exp":2}])
df=pd.concat([df,new_row],ignore_index=True)
new_row=pd.DataFrame([{"id":103,"Emp":"Clerk","Sal":10000,"Exp":3}])
df=pd.concat([df,new_row],ignore_index=True)
new_row=pd.DataFrame([{"id":104,"Emp":"President","Sal":1000000,"Exp":10}])
df=pd.concat([df,new_row],ignore_index=True)
print(df)
```

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Output:

Empty DataFrame
Columns: []
Index: []
<class 'pandas.core.frame.DataFrame'>
Empty DataFrame
Columns: [id, Emp, Sal, Exp]
Index: []

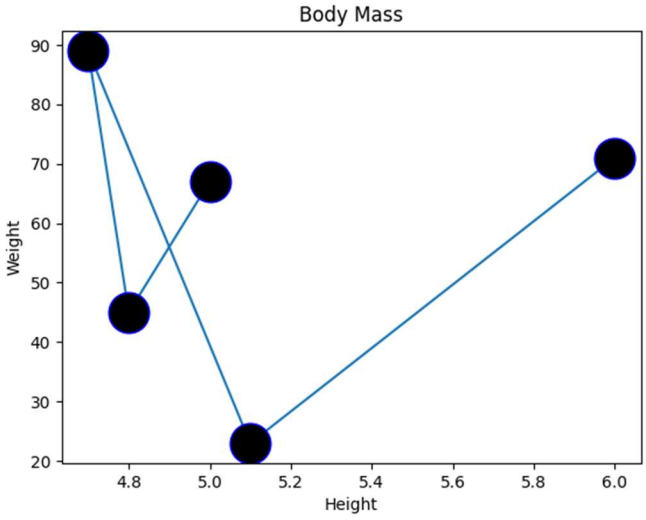
	id	Emp	Sal	Exp
0	101	Manager	100000	3
1	102	SalesPerson	60000	2
2	103	Clerk	10000	3
3	104	President	1000000	10

Implementation of Matplotlib Library:

PROGRAM:

```
import matplotlib.pyplot as plt
import numpy as np
x=np.array([5,4.8,4.7,5.1,6])
y=np.array([67, 45, 89, 23, 71])
plt.plot(x,y,marker='o',ms=25,mec='b',mfc='black')
plt.xlabel("Height")
plt.ylabel("Weight")
plt.title("Body Mass")
plt.show()
```

Output:



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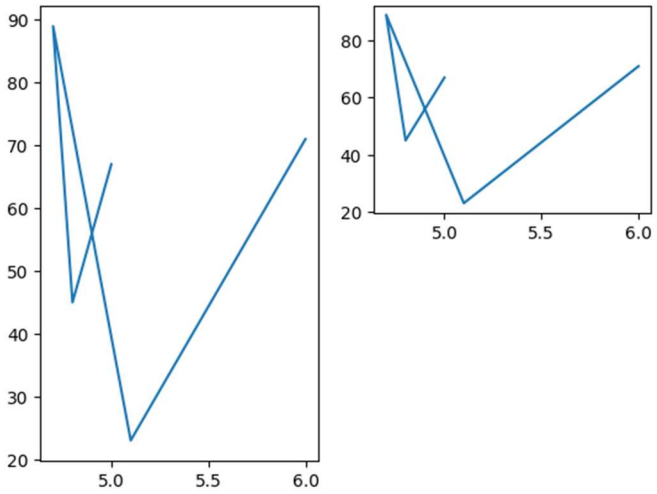
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Subplot using Matplotlib Library:

PROGRAM:

```
plt.subplot(1,2,1)
plt.plot(x,y)
plt.subplot(2,2,2)
plt.plot(x,y)
plt.show()
```

Output:

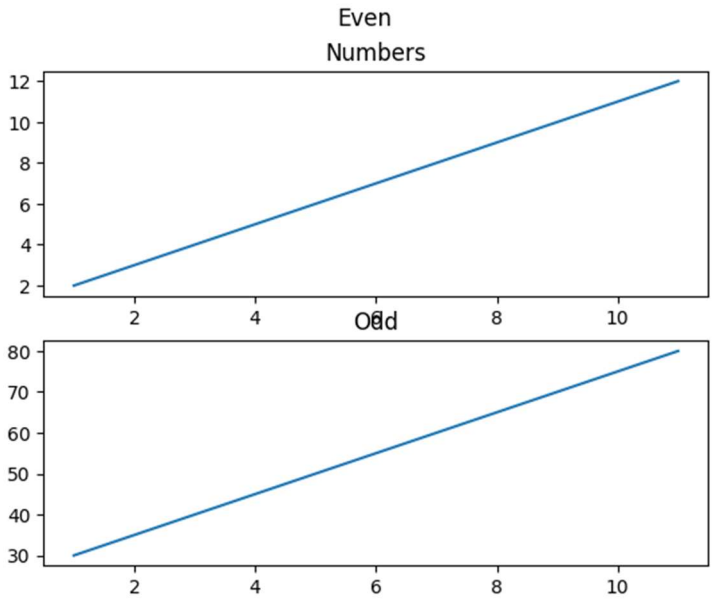


PROGRAM:

```
x=np.array([1,3,5,7,9,11])
y=np.array([2,4,6,8,10,12])
plt.subplot(2,1,1)
plt.plot(x,y)
plt.title("Numbers")
x1=np.array([1,3,5,7,9,11])
y1=np.array([30,40,50,60,70,80])
plt.subplot(2,1,2)
plt.plot(x1,y1)
plt.title("Odd")
plt.suptitle("Even")
plt.show()
```

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Output:

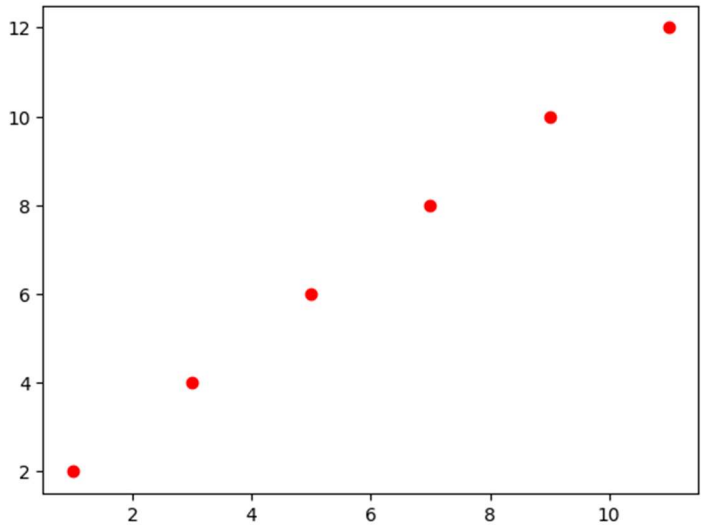


Scatter Plot using using Matplotlib Library:

PROGRAM:

```
x=np.array([1,3,5,7,9,11])
y=np.array([2,4,6,8,10,12])
plt.scatter(x,y,color='red')
plt.show()
```

Output:



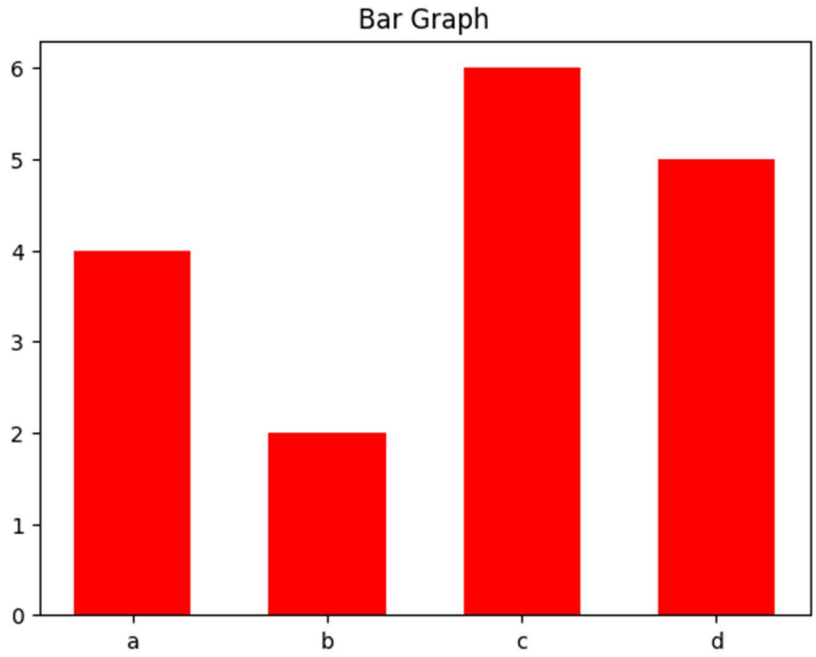
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BarGraph using Matplotlib Library:

PROGRAM:

```
x=np.array(["a","b","c","d"])
y=np.array([4,2,6,5])
plt.bar(x,y,color='red',width=0.6)
plt.title("Bar Graph")
plt.show()
```

Output:



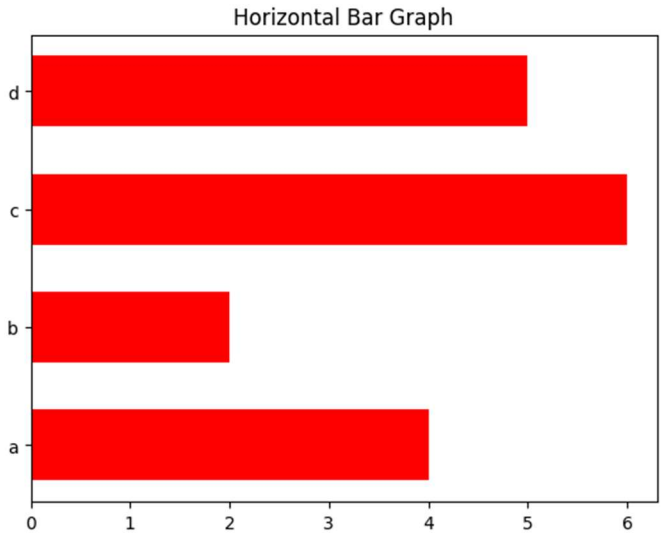
Horizontal BarGraph using Matplotlib Library:

PROGRAM:

```
x=np.array(["a","b","c","d"])
y=np.array([4,2,6,5])
plt.barh(x,y,color='red',height=0.6)
plt.title("Horizontal Bar Graph")
plt.show()
```

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Output:

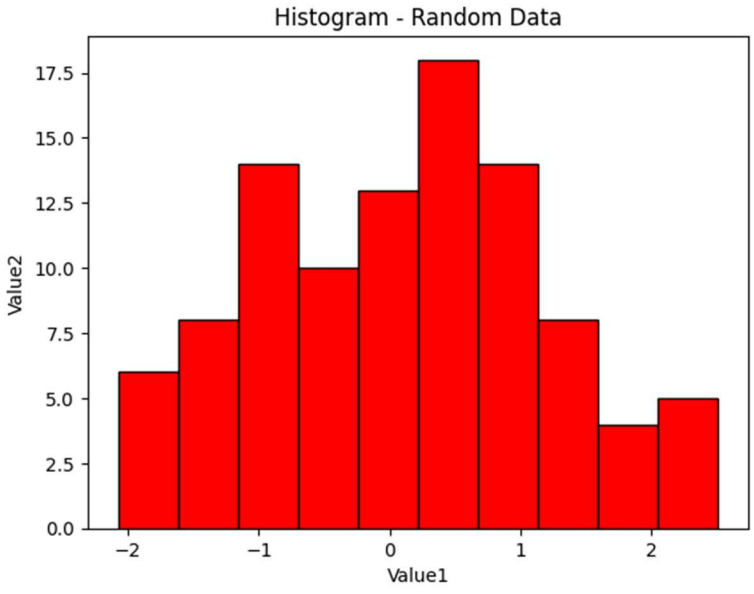


Histogram using Matplotlib Library:

PROGRAM:

```
data=np.random.randn(100)
plt.hist(data,bins=10,color='red',edgecolor='black',label="Random")
plt.title("Histogram - Random Data")
plt.xlabel("Value1")
plt.ylabel("Value2")
plt.show()
```

Output:



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Pie Chart using Matplotlib Library:

PROGRAM:

```
y=np.array([20,5,45,30])
myl=["ironman","captian","thor","hulk"]
plt.pie(y,labels=myl)
plt.show()
```

Output:

